11/13/2026: Expectation and Variance

CSCI 246: Discrete Structures

Textbook reference: Sec 34, Scheinerman

Graded Quiz Pickup

Quizzes are in the front of the room, grouped into four bins (A-G, H-L, M-R, S-Z) by last name. The quizzes are upside down with your last name on the back. Come find yours before, during, or after class. Only turn the quiz over if it's yours.

Today's Agenda

- Quiz
- Mini-lecture
- Group exercises

Quiz #10

1. Let (S, P) be the probability space for a single roll of a fair six-sided die. What is the sample space S? What is $P(s)$ for each outcome $s \in S$?
2. What is the probability of getting a **three of a kind** in a 5-card poker hand? (Note: A *three of a kind* refers to a poker hand where three cards have the same rank and the two other cards have different ranks. For example, a hand with three 10s, a 7, and a jack is a three-of-a-kind.)
3. *(Spinner Winner.)* Below are two spinners. When you spin A, you have a 50% chance of winning, and when you spin B, you have a 25% chance of winning. If you land on Win after a spin, you will win \$1 million. Which of the following options is your best course of action? (a) Spin A once, (b) Spin B twice, so you get two shots, or (c) It makes no difference. Justify your answer using a probabilistic argument.



Outline for Mini-Lecture:

- Overview
- Expectation
- Variance

Expectations and variances in real life (1/2)

Expectation tells you the average value of a random variable.

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Two companies offer LSAT preparation for law school. You may want to choose the company that offers a larger expected increase in LSAT scores.

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Variance tells you how much spread (or uncertainty) there is about a random variable.

Example of how variances are useful

Two companies offer LSAT preparation for law school. Both offer the same expected increase in LSAT score. However, they differ in the variability of increases in LSAT scores. (Company A has a larger spread in score boosts than company B.)

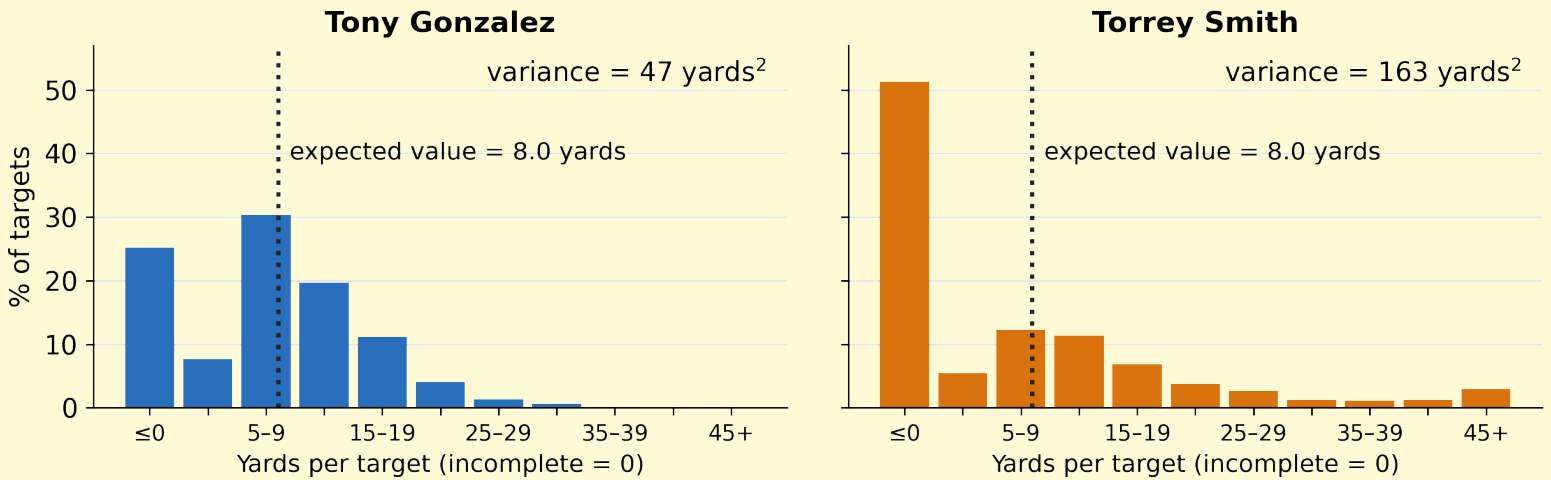
- If you want to minimize risk, you may want to choose company A.
- If you want to maximize potential gain, you may want to choose company B.

Expectations and variances in real life (2/2)

- **Expectation** tells you the average value of a random variable.
- **Variance** tells you how much spread (or uncertainty) there is about a random variable.

Example: Reasoning with expectations and variances

Tony Gonzalez and Torrey Smith both averaged about 8 yards per target, but Smith's boom-or-bust gains had about 3.5 times the variance.

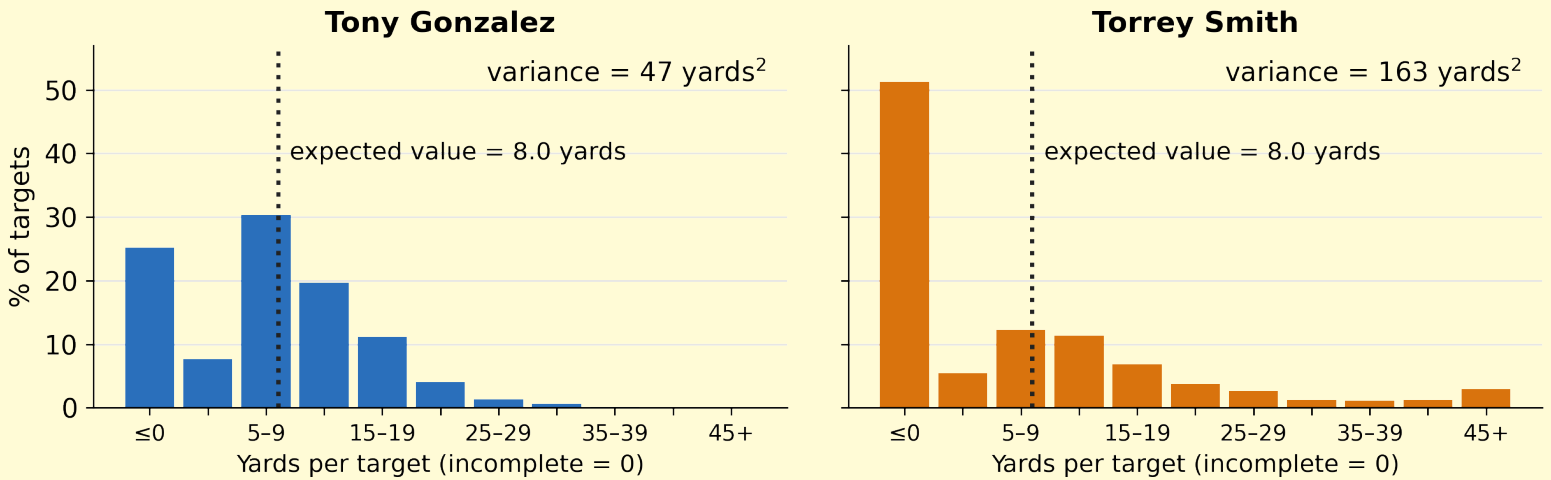


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Today, we'll practice computing these quantities from data.

Outline for Mini-Lecture:

- Overview
- Expectation
- Variance

Suppose we roll a pair of dice. What is the *expected value* of the sum?



Formally, let X be the sum of the number on the two dice. We must compute $E[X]$.

Computing An Expectation

Table 1: *Possible outcomes when rolling a pair of dice.* Each table entry gives the sum of the values on the two dice.

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$$E[X] = \sum_{s \in S} X(s)P(s)$$

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Application.

$$E[X] = 1{+}1 \cdot \frac{1}{36} + 1{+}2 \cdot \frac{1}{36} + 2{+}1 \cdot \frac{1}{36} + 3{+}1 \cdot \frac{1}{36} + 2{+}2 \cdot \frac{1}{36} + 1{+}3 \cdot \frac{1}{36} + \ldots + 6{+}6 \cdot \frac{1}{36}$$

$$= 2 \cdot \frac{1}{36} + 3 \cdot \frac{1}{36} + 3 \cdot \frac{1}{36} + 4 \cdot \frac{1}{36} + 4 \cdot \frac{1}{36} + 4 \cdot \frac{1}{36} + \ldots + 12 \cdot \frac{1}{36}$$

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Remark. We must sum over **many** (36) terms.

Computing An Expectation

Table 2: *Possible outcomes when rolling a pair of dice.* Each table entry gives the sum of the values on the two dice.

Dice Values	1	2	3	4	5	6
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$$E[X] = \sum_{a \in \mathbb{R}} a \, P(X = a)$$

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Remark. We must sum over **fewer** (12) terms.

Computing An Expectation

Table 3: *Possible outcomes when rolling a pair of dice.* Each table entry gives the sum of the values on the two dice.

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Method 3. Use linearity of expectation.

$$E[X] = E[D_1] + E[D_2]$$

where D_n is the value on the n-th die.

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Application. We have

$$E[X] = E[D_1] + E[D_2] = 3.5 + 3.5 = 7.$$

since, by the previous methods, each $E[D_n] = \frac{1}{6} \cdot (1 + 2 + 3 + 4 + 5 + 6) = 3.5$.

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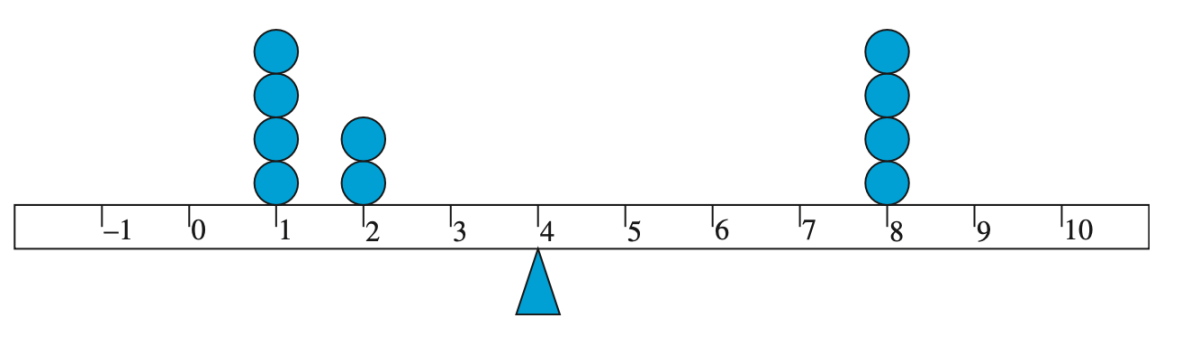
Remark. Here we can sum over **even fewer** (2 or 7) terms. (Note: we can only use Method 3 in cases where we can express X as a sum of other RVs).

Expectation as center of probability mass

Consider a probability space (S, P) where $S = \{1, 2, \ldots, 10\}$ and $P(s) = \frac{1}{10}$ for all outcomes $s \in S$. Define a random variable X as below:

s	$X(s)$	s	$X(s)$
1	1	6	2
2	1	7	8
3	1	8	8
4	1	9	8
5	2	10	8

Now we make a seesaw, placing a weight $P(X = a)$ for each outcome a that the random variable can take on:



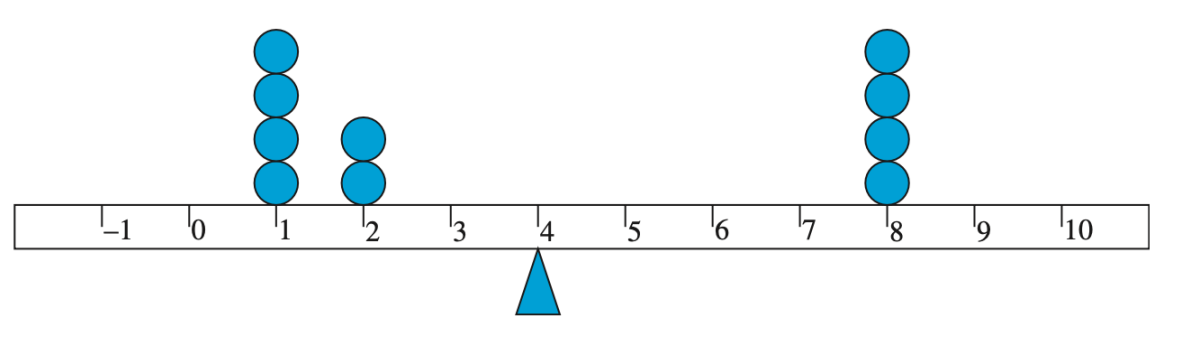
What is the expected value?

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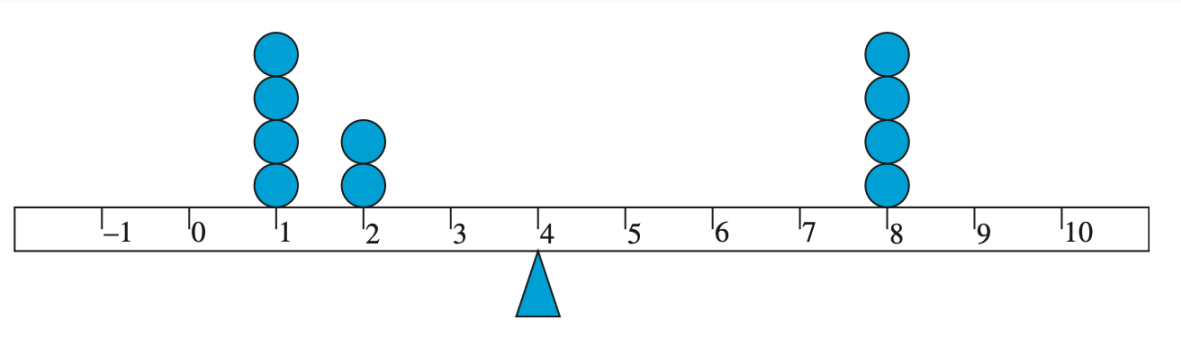
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What is the expected value? The **balancing point** (▲) of the seesaw!

Expectation as center of probability mass

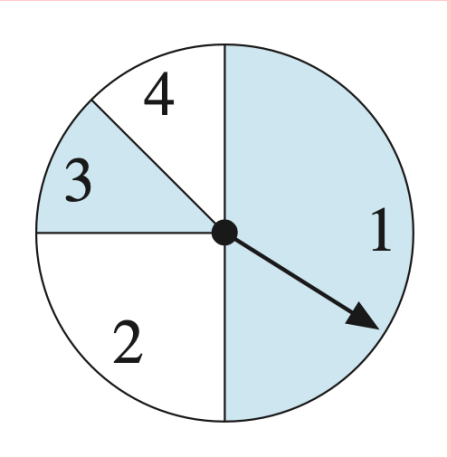


To be concrete, the expected value for this problem is given by

$$\begin{aligned} E[X] &= \sum_{a \in \mathbb{R}} a\, P(X = a) \\ &= 1 \times 0.4 + 2 \times 0.2 + 8 \times 0.4 \\ &= 4 \end{aligned}$$

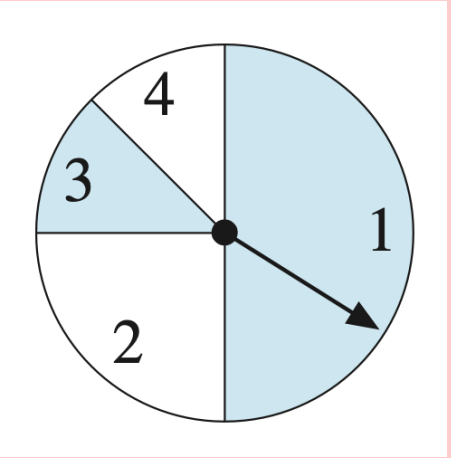
Practice Reading Quiz (Expectation)

Consider the spinner shown in the figure. Suppose that the likelihood of landing in each region is proportional to the area of the region. Let X be the number that appears on the spinner. Compute the expected value of X.



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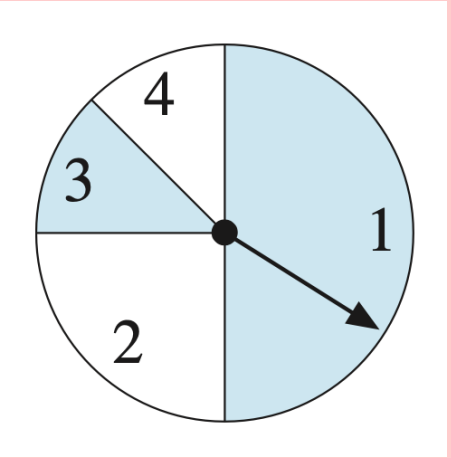


Solution.

$$E[X] = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 4 = \frac{15}{8}.$$

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Remark. The expected value is a **weighted average**, where the weights are probabilities.

Outline for Mini-Lecture:

- Overview
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- Variance

Variance as measure of spread

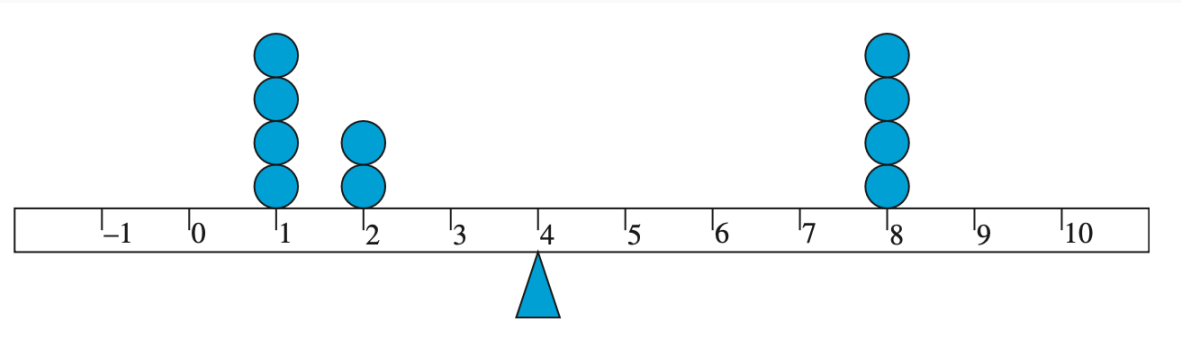
Definition

Let X be a real-valued random variable on a probability space (S, P). Let $\mu := E[X]$. Then the **variance** of X is

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mu)^2\right]$$

Remark

The variance is the **expected squared deviation** from the mean.



Variance as measure of spread

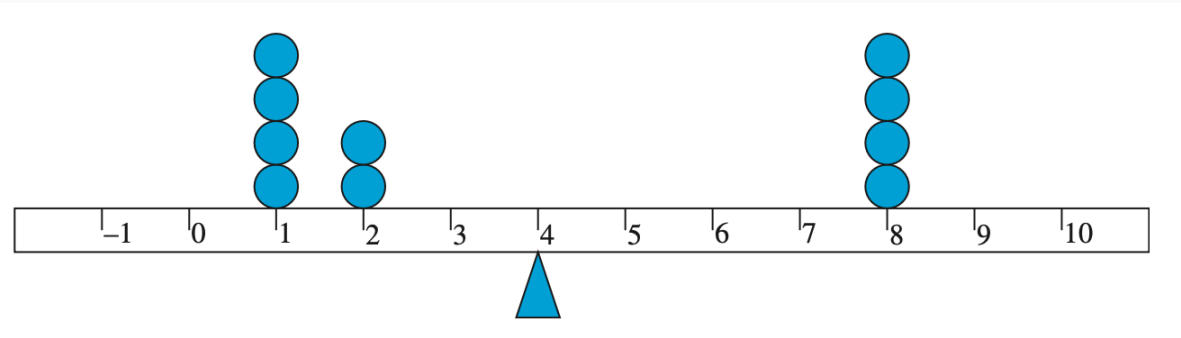
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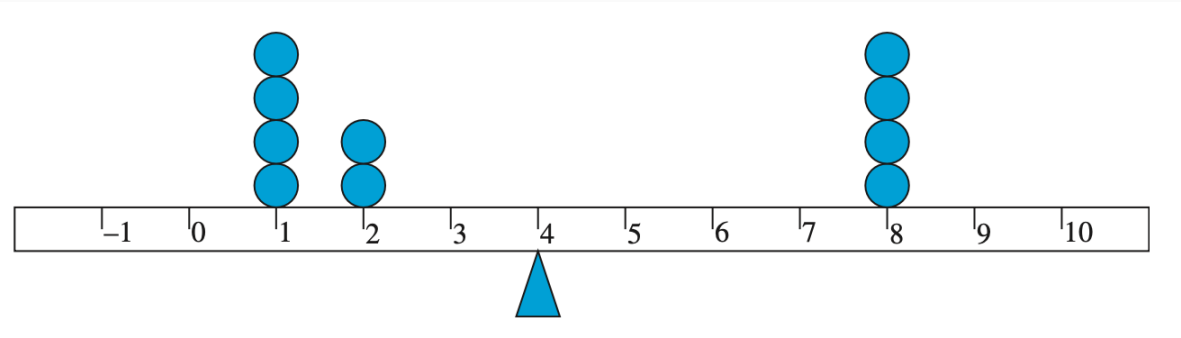
Remark

The variance is the **expected squared deviation** from the mean.



Poll. Can anybody here explain how to compute the variance for this problem?

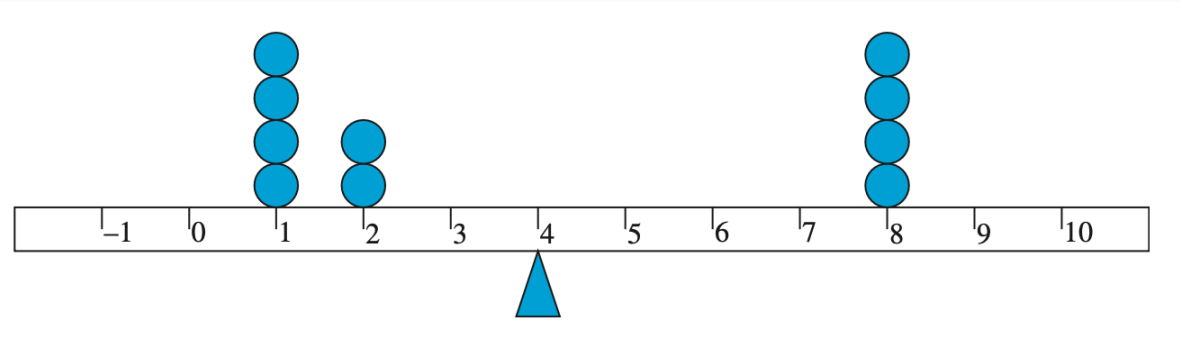
Expectation vs. variance



Here, the **expected value** $E[X]$ (a.k.a. the mean μ) is given by

$$\begin{aligned} E[X] &= \sum_{a \in \mathbb{R}} a \; P(X = a) \\ &= 1 \times 0.4 + 2 \times 0.2 + 8 \times 0.4 \\ &= 4 \end{aligned}$$

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s The **variance** $Var(X)$ (i.e. the expected squared deviation) is given by

$$\begin{aligned} Var(X) &= \sum_{a \in \mathbb{R}} (a - \mu)^2 \; P(X = a) \\ &= (1\text{-}4)\text{^}2 \times 0.4 + (2\text{-}4)\text{^}2 \times 0.2 + (8\text{-}4)\text{^}2 \times 0.4 \\ &= 9 \times 0.4 + 4 \times 0.2 + 16 \times 0.4 \\ &= 10.8 \end{aligned}$$

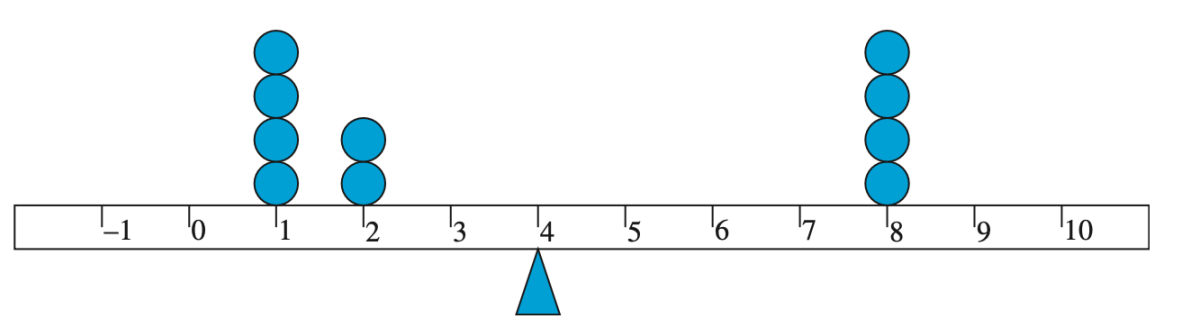
Remark

When computing the variance, we can use

$$\mathrm{Var}(X) = \sum_{s \in S} [X(s) - \mu]^2 \; P(s)$$

or

$$= \sum_{a \in \mathbb{R}} (a - \mu)^2 \; P(X = a)$$



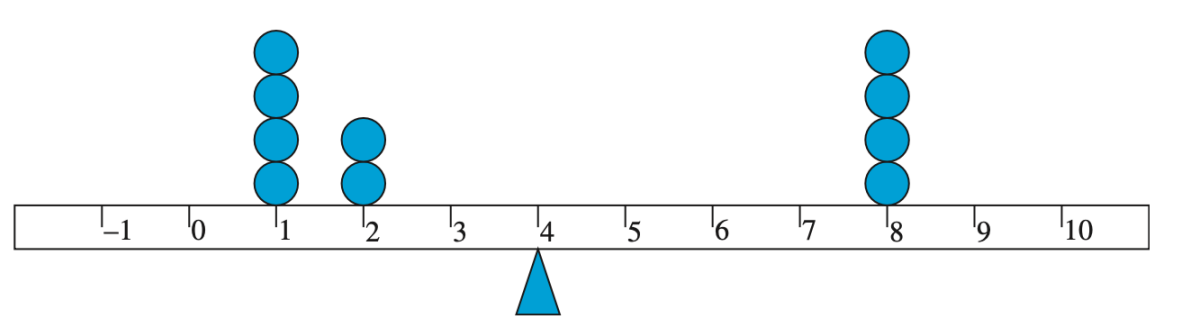
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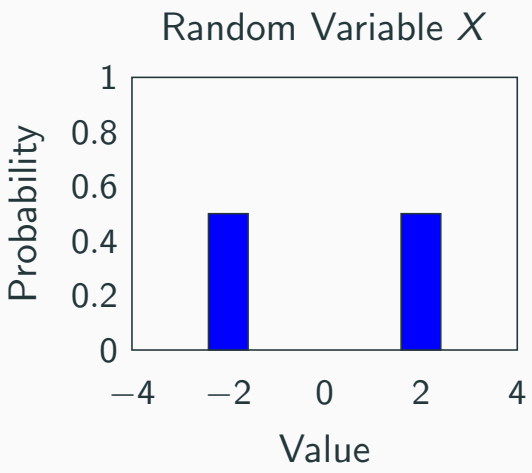
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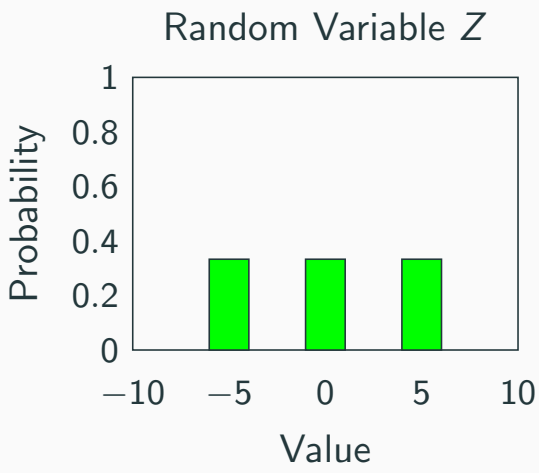
Poll. Can anybody explain how two apply these different methods to this problem?

Three Random Variables

Poll. These three random variables all have the same mean. Which has the largest variance?







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Solution. Z has the largest variance. (See textbook for more info.)

Group exercises

Random group assignments

Ashley, Isabelle: 10
Bailey, Tanna: 11
Blaisdell, Olyver: 1
Brown, Ashton: 10
Cardenas, Dominic: 4
Cutler, George: 1
Fabik, Roland: 11
Fellows, Audrey: 3
Fitzgerald, Ryan: 9
Fuller, Ryan: 10
Griffen, Clara: 12
Grutsch, Tanner: 8
Hager, Nathan: 3
Horner, Jayden: 4
Johnson, Jaret: 7
Kintzel, Samantha: 9
Lameres, Alexis: 8
Lee, Holden: 6
Mayala, Reece: 11
McLean, Connor: 5
Mcbride, Daniel: 2
Miller, Maverick: 5
Moss, Emily: 2
Mousseau, Matthew: 7
Neuman, Evan: 6
Putnam, John: 7
Ramos, Vydel: 3
Rathbun, Jedidiah: 8
Roberts, Matthew: 1
Roller, Nathan: 5
Ryan, Otis: 12
Shevtsov, Timothy: 4
Stensland, Emma: 6
Thiessen, Logan: 2
Thompson, Jaiden: 9
Vath, Ethan: 12

Group exercises

1. *Simplified stock market.* Suppose there are three kinds of days: `GOOD`, `GREAT`, and `ROTTEN`. The following chart gives the frequency of each of these types of days and the effect on the price (in dollars) of a certain stock that day.

Type of day	Frequency	Change in stock value
`GOOD`	60%	+2
`GREAT`	10%	+5
`ROTTEN`	30%	-4

Let X_n be the change in the value of the stock after n consecutive days.

   a. Find $E[X_1]$, the expected change in stock price after one day.
   b. Find $Var[X_1]$, the variance in the change in stock price after one day.
   c. Find $E[X_5]$.
   d. Find $Var[X_5]$.

2. A pair of 25-sided dice is rolled. Let X be the sum of the two numbers and Y be the product. Find $E[X]$ and $E[Y]$.



3. The term *expected value* can be a bit deceiving. Sometimes it is not what someone might expect! (a) Give an example of a random variable X where $E[X] = 1$ but $P(X = 1) = 0$. (b) Give an example of a random variable X where $E[X] < 0$ but the probability that X is positive is nearly 100%.

Solution to group exercise #1a

Problem & Solution. *Simplified stock market.* Suppose there are three kinds of days: `GOOD`, `GREAT`, and `ROTTEN`. The following chart gives the frequency of each of these types of days and the effect on the price (in dollars) of a certain stock that day.

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`GOOD`	60%	+2
`GREAT`	10%	+5
`ROTTEN`	30%	-4

Let X_n be the change in the value of the stock after n consecutive days.

a. Find $E[X_1]$, the expected change in stock price after one day.

$$E[X_1] = \sum_{s \in S} X_1(s)P(s)$$
$$= (+2)(0.6) + (+5)(0.1) + (-4)(.30) = 0.5$$

We expect the stock value to rise \$0.50 after one day.

Solution to group exercise #1b

Problem & Solution. *Simplified stock market.* Suppose there are three kinds of days: `GOOD`, `GREAT`, and `ROTTEN`. The following chart gives the frequency of each of these types of days and the effect on the price (in dollars) of a certain stock that day.

Type of day	Frequency	Change in stock value
`GOOD`	60%	+2
`GREAT`	10%	+5
`ROTTEN`	30%	-4

Let X_n be the change in the value of the stock after n consecutive days.

b. Find $Var[X_1]$, the variance in the change in stock price after one day.

$$Var[X_1] = \sum_{s \in S} (X_1(s) - \mu)^2 P(s) \qquad \text{(where } \mu := \mathbb{E}[X_1]\text{)}$$

Since we know from part (a) that $\mu = 0.5$,

$$= \sum_{s \in S} (X_1(s) - 0.5)^2 P(s)$$

$$= (2 - 0.5)^2 \times .6 + (5 - 0.5)^2 \times .3 + (-4 - 0.5)^2 \times .10$$

$$= 9.45$$

Solution to group exercise #1b – Alternate solution.

Problem & Solution. *Simplified stock market.* Suppose there are three kinds of days: `GOOD`, `GREAT`, and `ROTTEN`. The following chart gives the frequency of each of these types of days and the effect on the price (in dollars) of a certain stock that day.

Type of day	Frequency	Change in stock value
`GOOD`	60%	+2
`GREAT`	10%	+5
`ROTTEN`	30%	-4

Let X_n be the change in the value of the stock after n consecutive days.

b. Find $Var[X_1]$, the variance in the change in stock price after one day.

$$Var[X_1] = E[X_1^2] - E[X_1]^2$$

For the first summand,

$$\begin{aligned} E[X_1^2] &= \sum_{s \in S} X_1(s)P(s) \\ &= (+2)^2 \times .6 + (+5)^2 \times .10 + (-4)^2 \times .30 = 0.5 \\ &= 4 \times .6 + 25 \times .10 + 16 \times .30 = 9.7 \end{aligned}$$

For the second summand, we know from part (a) that $\mathbb{E}[X_1] = 0.5$. Hence $\mathbb{E}[X_1]^2 = 0.25$. Putting this together, we have

$$Var[X_1] = E[X_1^2] - E[X_1]^2 = 9.7 - 0.25 = 9.45.$$

Solution to group exercise #1c

Problem & Solution. *Simplified stock market.* Suppose there are three kinds of days: `GOOD`, `GREAT`, and `ROTTEN`. The following chart gives the frequency of each of these types of days and the effect on the price (in dollars) of a certain stock that day.

Type of day	Frequency	Change in stock value
`GOOD`	60%	+2
`GREAT`	10%	+5
`ROTTEN`	30%	-4

Let X_n be the change in the value of the stock after n consecutive days.

c. Find $E[X_5]$, the expected change in stock price after five days. The analysis is simplest if we write $X_5 = Y_1 + Y_2 + \ldots + Y_5$, where Y_j is the change in stock price on the j-th day.

Now, by linearity of expectation

$$E[X_5] = E[Y_1] + E[Y_2] + \ldots + E[Y_5].$$

Each summand is given by

$$E[Y_j] = (+2)(0.6) + (+5)(0.1) + (-4)(.30) = 0.5,$$

as was solved in part (a).

Hence,

$$E[X_5] = 5(0.5) = 2.5$$

We expect the stock value to rise \$2.50 after 5 days.

Solution to group exercise #1d

Problem. Find $Var[X_5]$, the variance of the change in stock price after five days.

Solution. We write

$$Var[X_5] = E[X_5^2] - E[X_5]^2$$

For the second summand, we know from part (c) that $E[X_5] = 2.5$, so $E[X_5]^2 = 2.5^2 = 6.25$.

For the first summand, we again write $X_5 = Y_1 + Y_2 + \ldots + Y_5$, where Y_j is the change in stock price on the j-th day. We use the fact that the Y_j's are independent.

$$\begin{aligned}
E[X_5^2] &= E[(Y_1 + Y_2 + \ldots + Y_5)^2] \\
&= E[Y_1^2 + \ldots + Y_5^2 + 2Y_1Y_2 + \ldots + 2Y_4Y_5] && \text{(expanding the sum of squares)} \\
&= 5E[Y_1^2] + 20E[Y_1Y_2] && \text{(since } Y_j\text{'s are identical)} \\
&= 5E[Y_1^2] + 20E[Y_1]E[Y_2] && \text{(by independence)} \\
&= (5)(9.7) + 20(0.5)^2 && \text{(using previous parts)} \\
&= 53.5
\end{aligned}$$

Hence,

$$Var[X_5] = E[X_5^2] - E[X_5]^2 = 53.5 - 6.25 = 47.25$$

Remark. This problem is much simpler if we use a fact (not presented in the text) that

$$Var(X_5) = Var(Y_1) + \ldots Var(Y_5)$$

because the Y_j's are independent.

Solution to group exercise #2

Problem. A pair of 25-sided dice is rolled. Let X be the sum of the two numbers and Y be the product. Find $E[X]$ and $E[Y]$.

Solution. Let X_i be the value of the die on the i-th roll, where $i \in \{1, 2\}$.
Now $X = X_1 + X_2$. So by linearity of expectation, $E[X] = E[X_1] + E[X_2]$. Each X_i satisfies

$$\begin{aligned} E[X_i] &= \sum_{s \in S} X(s)P(s) \\ &= (1 + \ldots + 25)\frac{1}{25} \\ &= \frac{(25)(26)}{2}\frac{1}{25} \\ &= 13 \end{aligned}$$

Hence

$$E[X] = E[X_1] + E[X_2] = 13 + 13 = 26$$

So on average, the sum of the values on the two dice will be 26.
Moreover $Y = X_1 X_2$. By independence, $E[Y] = E[X_1]E[X_2]$. So the expected product of the two rolls is

$$E[Y] = E[X_1]E[X_2] = (13)(13) = 169.$$

So on average, the product of the values on the two dice will be 169.

Solution to group exercise #3

Problem. The term *expected value* can be a bit deceiving. Sometimes it is not what someone might expect! (a) Give an example of a random variable X where $E[X] = 1$ but $P(X = 1) = 0$. (b) Give an example of a random variable X where $E[X] < 0$ but the probability that X is positive is nearly 100%.

Solution.

a. Let X be a random variable that takes on the value $X = 2$ with probability $\frac{1}{2}$ and the value $X = 0$ with probability $\frac{1}{2}$. Then

$$E[X] = \sum_{a \in \mathbb{R}} a\, P(X = a) = \left(\frac{1}{2}\right)(0) + \left(\frac{1}{2}\right)(2) = 1.$$

But $P(X = 1) = 0$.

b. Let X be a random variable that takes on the value $X = -1,000,000$ with probability 0.001 and the value $X = 1$ with probability 0.999. Then

$$E[X] = \sum_{a \in \mathbb{R}} a\, P(X = a) = (.001)(-1,000,000) + (.999)(1) = -999.001.$$

So $E[X] < 0$; that is, the expected value of the random variable is negative. But

$$P(X > 0) = P(X = 1) = 0.999$$

So the probability that X is positive is almost 100%.